Literature status: rank inequalities and 2×2 orbit degeneration

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Status. Literature already answered (partial subcase). The independent proof below is valid but is not claimed as new.

1 Source question

Derksen, Klep, Makam, and Volčič [1] ask at the end of Section 6 whether their rank inequalities are equivalent to stable orbit degeneration. In their notation, for $A, B \in \mathrm{Mat}_n(k)^m$, condition (b) is

$$\operatorname{rank}\left(I \otimes T_0 + \sum_i A_i \otimes T_i\right) \leq \operatorname{rank}\left(I \otimes T_0 + \sum_i B_i \otimes T_i\right) \tag{R}$$

for all square affine linear matrix pencils, while (a') says that for some ℓ, $A \oplus 0_\ell$ lies in the closure of the similarity orbit of $B \oplus 0_\ell$. This is related to the open comparison between the hom order and virtual degeneration order.

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> and so $A \oplus 0_1 = \lim_{t\to 0} P_t(B \oplus 0_1)P_t^{-1}$. Therefore (b) holds for $A \oplus 0_1$ and $B \oplus 0_1$, and consequently also for A and B.
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> The above example indicates that (a) above should be replaced by
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> (a') for some $\ell \in \mathbb{N}$, $A \oplus 0_\ell$ lies in the closure of the $\mathrm{GL}_{n+\ell}$-orbit of $B \oplus 0_\ell$.
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> The problem of equivalence of (a') and (b) has a counterpart in representation theory. There it is the open question of whether the virtual degeneration order and the hom order are equivalent [Sma08, Section 5] (more precisely, virtual degeneration allows for the zero tuple 0_ℓ in (a') to be replaced by an arbitrary $C \in \mathrm{Mat}_\ell^m$).
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> 7. Algorithms
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> In this section we give algorithms pertaining to the main results of the paper. A deterministic polynomial time algorithm for testing orbit equivalence under similarity

2 Idea of the proof

The source's matricization identifies (R) with inequalities $\dim \mathrm{Hom}(X, M_A) \geq \dim \mathrm{Hom}(X, M_B)$ for all finite-dimensional modules X. In dimension two there is almost no room. A nonsemisimple module is a nonsplit extension of two one-dimensional simples and has a unique proper nonzero submodule. Testing first with that submodule and then with the whole extension forces M_A either to be the same extension or to contain both simple factors as submodules. The latter means that M_A is the split extension, which is visibly a limit of the nonsplit one. Thus stabilization is unnecessary in this dimension. The general question for $n \geq 3$ remains open.

3 Later literature correction

Domokos and Miklósi [2] explicitly study the same Hom-order and degeneration-order comparison, citing the source paper. Their Corollary 3.3 proves that the two orders coincide for all 2×2 matrix tuples, and their Theorem 5.4 proves the same for nilpotent 3×3 tuples. Thus the theorem below is a literature-known partial subcase. The later paper also records Carlson's dimension-four counterexample to the *unstabilized* implication. It does not settle the source's stabilized/virtual-degeneration question in general.

4 Independent proof

Let k be algebraically closed and let an m-tuple act on k^n as a finite-dimensional module over the free algebra $k\langle x_1, \ldots, x_m \rangle$.

Theorem 1. *For $A, B \in \mathrm{Mat}_2(k)^m$, condition* (R) *holds if and only if $A \in \overline{B^{\mathrm{GL}_2}}$. Consequently (a') and (b) in the source are equivalent for $n = 2$, and one may take $\ell = 0$.*

Proof. Orbit degeneration implies (R) by lower semicontinuity of matrix rank. For the converse, let $M = M_A$ and $N = M_B$. Lemma 3.1 and the padding argument in [1] translate (R) into

$$\dim \mathrm{Hom}(X, M) \geq \dim \mathrm{Hom}(X, N) \tag{H}$$

for every finite-dimensional module X.

Suppose first that N is semisimple. If N is simple, (H) with $X = N$ gives a nonzero map $N \to M$; it is injective and, dimensions being equal, an isomorphism. If $N = S \oplus T$ for distinct one-dimensional simples, (H) with $X = S, T$ puts a copy of each in the socle of M, so $M \cong S \oplus T$. If $N = S \oplus S$, then $\dim \mathrm{Hom}(S, M) \geq 2$; the common S-eigenspace is all of M, again giving $M \cong N$.

It remains that N is a nonsplit extension

$$0 \longrightarrow S \longrightarrow N \longrightarrow T \longrightarrow 0 \tag{E}$$

of one-dimensional simples. It has a unique proper nonzero submodule, namely S. Moreover, $\dim \mathrm{Hom}(S, N) = 1$, so (H) puts a copy of S in M. Moreover,

$$\dim \mathrm{Hom}(N, M) \geq \dim \mathrm{End}(N).$$

If one of these maps $N \to M$ has rank two, then $M \cong N$.

Assume otherwise that every such map has rank at most one. When $S \ncong T$, the endomorphism algebra of the nonsplit extension is one-dimensional, so there is a nonzero rank-one map $N \to M$. Its kernel must be the unique submodule S of N, and hence its image is a copy of T. Thus M contains the two distinct simple submodules S and T, and $M \cong S \oplus T$.

When $S \cong T$, write, in a suitable basis, $B_i = \alpha_i I + b_i E_{12}$, with some $b_i \neq 0$. Then $\mathrm{End}(N) = \mathrm{span}\{I, E_{12}\}$ has dimension two. Every rank-one map $N \to M$ kills the unique submodule S and factors through the fixed quotient $N \twoheadrightarrow S$. Hence (H) forces $\dim \mathrm{Hom}(S, M) \geq 2$. The common S-eigenspace is therefore all of M, so $M \cong S \oplus S$.

In either nonisomorphic case $M \cong S \oplus T$. Finally, writing $B_i = \left(\begin{smallmatrix} \alpha_i & b_i \\ 0 & \beta_i \end{smallmatrix}\right)$, conjugation by $\mathrm{diag}(t, 1)$ multiplies every b_i by t. Letting $t \to 0$ shows that $S \oplus T$ lies in $\overline{N^{\mathrm{GL}_2}}$. Thus $A \in \overline{B^{\mathrm{GL}_2}}$ in every case. $\square$

5 Verification, limitations, and provenance

The proof uses no computation. Its cases exhaust all two-dimensional modules over an algebraically closed field: simple, semisimple of length two, or a nonsplit length-two extension. The theorem does not address non-algebraically closed fields or general $n \geq 3$. The initial search missed [2]; this revised packet corrects that error. No novelty claim is made. The general stabilized question remains open.

Human-review recommendation. Check especially that the source's rectangular Hom matricization is padded without changing the direction of (R), and the dimension-two self-extension argument.

References

[1] H. Derksen, I. Klep, V. Makam, J. Volčič, *Ranks of linear matrix pencils separate simultaneous similarity orbits*, arXiv:2109.09418, especially Lemma 3.1 and Section 6.

[2] M. Domokos, B. Miklósi, *Degeneration order of* 3×3 *nilpotent matrix tuples*, arXiv:2604.22609, especially Corollary 3.3 and Theorem 5.4.